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Numerical study of heat transfer in underground power cable system

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Abstract

Underground power cable system has been widely used as an effective transmission line. This paper presents the thermal analysis of the two formations (Flat and Trefoil) of buried system. Temperature-dependent thermal conductivity of soil was considered in the research and the Fluidized Thermal Backfill (FTB) was applied to increase the ability of heat dissipation of power cable system. Further, the influences of heat transfer performance and ampacity on power cable systems were studied when FTB was applied in two formations. Firstly, it is found that the geometric parameters have significant effects on the heat transfer performance of underground power cable system. Secondly, considering the temperature-dependent thermal conductivity, the maximum temperature of cables will rise rapidly, the limited temperature is easy to reach. The application of FTB can enhance the heat transfer of the whole system, especially for trefoil formation. Finally, the analysis of ampacity shows that the FTB will ensure the higher loading current in a security condition with the temperature-dependent thermal conductivity.

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Keywords: Power cable system, thermal conductivity; ampacity

1. Introduction

The operation security of power cable system attracted a broad scientific attention. In most instance, the maximum electric current, which is ampacity, is adopted as large as possible for a greater profit. For safety consideration, the electric current is limited by the temperature of conductor for the Joule's law. The recommended

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optimum temperature is 65°C and the limited temperature is 90°C [1]. Therefore, heat dissipation process from the cable conductor to the surrounding soil plays a crucial role in evaluating the performance of buried cable system. The traditional method [2] calculate the thermal resistance between the underground power system and the external environment based on IEC standard. This method assumes that the soil is homogeneous medium with constant thermal conductivity. However, the process of heat transfer is more complex and the thermal conductivity of actual soil is dependent on other factors like temperature, porosity, liquid-vapor transport [3], which will lead to the errors of ampacity and temperature distribution compared with real condition. In fact, dry zones are usually formed [7], which will lead to the sharp increase of temperature of whole power cable system. In recent years, numerical simulations and experiments are gradually used to study the underground power cables systems [4–6]. They used different analysed methods to study heat transfer of underground power cables system. However, most of them only focused on the heat transfer performance in Flat formation and ignored the effect of the temperature on thermal conductivity of backfill material.

In this paper, the thermal conductivity of the native soil (NS) and FTB dependent on temperature were considered. The effects of geometrical parameters [8] on heat transfer performance of two installation formation (Flat and Trefoil) for buried power cable were investigated numerically with COMSOL software. Further, the influences of heat transfer performance and ampacity on power cable systems were studied when FTB was applied in two formations.

2. Geometrical and physical models

2.1. Geometrical model

In our study, two typical configurations including Flat formation and Trefoil formation shown in Fig. 1 is buried in native soil with length $L=20\text{m}$ and width $H=10\text{m}$. In Fig. 1(a), three power cables are buried at the depth h from 1m to 2m in Flat formation, the spacing between the subsequent cable cores d is from 0.2m to 0.6m. As for the Trefoil formation (Fig. 1b), the parameter h is the point we focus on. The distance between the cable and the bottom edge is 0.2m. Fig. 1c shows the power cable cross-section. Symbols r_c , r_{ins} , r_{sh} and r_j denote, respectively, the cable conductor, insulation, sheath, and jacket outer radiuses. The thicknesses of power cable layout materials are 24.8mm, 55.3mm, 61.7mm, and 66.8mm. In present study, the first of the considered laying case is that the power transmission line is situated directly in the NS. The second laying case is that power cables are buried in FTB bedding of rectangular cross-section, further, the FTB is buried in the NS.

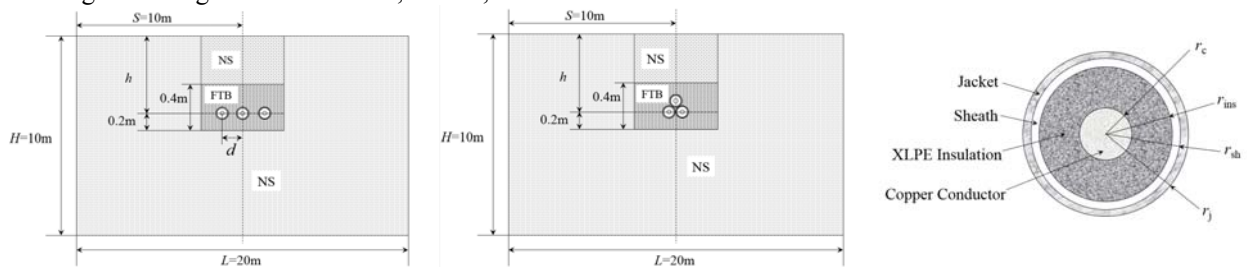


Fig. 1. (a) Flat formation; (b) Trefoil formation; (c) The model of power cable

2.2. Physical model

In practice, the cable line length is considerable, thus only the cross-section of soil, FTB and underground power cables is considered, which is two-dimensional. The temperature at the top edge is kept as 20°C and the boundary conditions for other edges are adiabatic. The two-dimensional steady-state heat conduction equation are as follows:

$$\frac{\partial}{\partial x} \left[k(T) \frac{\partial T}{\partial x} \right] + \frac{\partial}{\partial y} \left[k(T) \frac{\partial T}{\partial y} \right] = -q_v(T_c) \quad (1)$$

where T denotes the temperature of the underground power system, $k(T)$ is the thermal conductivity dependent on temperature in backfill field and is constant in the power cable, $q_v(T_c)$ is the heat source produced by the conductor, which is calculated by the following equation:

$$q_v(T) = |\vec{J}|^2 / \sigma \quad (2)$$

where σ is the conductivity of conductor, which is dependent on temperature. $\vec{J}$ is the current density. The $\vec{J}$ and σ are calculated as:

$$\sigma = \frac{1}{\rho_0(1 + \alpha(T - T_{ref}))} \quad (3)$$

$$\vec{J} = \vec{J}_e + \vec{J}_s \quad (4)$$

$$\begin{cases} \vec{J}_e = -j\omega\sigma\vec{A} \\ \vec{J}_s = -\sigma\nabla\phi \end{cases} \quad (5)$$

where $\vec{J}_e$ is induction current density, $\vec{J}_s$ is the source current density, ρ_0 and α are the reference resistivity and resistance temperature coefficient, respectively, both given at the reference temperature $T_{ref}=20^\circ\text{C}$. For described computational cases (copper conductor) $\rho_0=1.72\text{e-}8\Omega\cdot\text{m}$, $\alpha=0.0039$. In this paper, the loading current is considered as 1145A.

Considering the influence of temperature on the soil properties, Joule's heat generated in the power cable conductor may cause the formation of dry zone in native soil and the increase of thermal conductivity under loading condition. Further, the dry zone nearby the cables will lead to a deterioration of heat dissipation. The varied thermal conductivity can be described by following equation [9]:

$$k(T) = k_{dry} + (k_{wet} - k_{dry}) \cdot \exp[-a_1[(T - T_{ref}) / (a_2 \cdot T_{max,p})]^2] \quad (6)$$

where k_{dry} and k_{wet} are the thermal conductivities in dry and wet. $T_{max,p}$ is the cable maximum allowable operating temperature, which is 90°C . The a_1 and a_2 are coefficients defined as:

$$a_1 = T_{max,p} / T_{ref} \quad (7)$$

$$a_2 = 1 - (1 / a_1) \quad (8)$$

3. Results and discussion

3.1. Model validation and grid independency study

The numerical calculation in the present study is validated by simulating a direct buried three power cables system, the current is 1145A, and the T_m is compared with the result of literature [1]. The change of the maximum temperature with the burial depth is shown in Fig. 2(a). The distribution of temperature for symmetry axis is shown in Fig.2(b). The maximum error in Fig. 2(a) and Fig. 2(b) are 1.23% and 3.44%, respectively. The comparisons of the results reveal that the present numerical method is reasonable and acceptable.

In order to obtain a grid independent solution, four different grids (12185, 16337, 19346, 28394) are chosen to calculate our numerical model (without FTB, $k(T)=\text{const}$, $I=1145\text{A}$), the comparison shows that there is almost no

difference between the result of the mesh with grid number of 19346 and 28394. So the mesh with grid number of 19346 is employed for calculation.



Fig. 2. (a) Comparison of T_m with the change of burial depth; (b) Comparison of distribution of temperature along the symmetry axis

3.2. The effects of buried depth and spacing of cables with constant thermal conductivity

Fig. 3 shows the performance of T_m in Flat formation and Trefoil formation with constant thermal conductivity. The first laying case is adopted. Fig. 3(a) depicts the variation of maximum temperature with h . It shows that the T_m of Flat formation is lower than Trefoil formation, both of them increase with the increase of burial depth. The T_m of two formations are all lower than limiting temperature. The T_m of Trefoil will reach the optimum operation temperature when the buried depth exceeds 1.75m. Fig. 3(b) shows the influence of d on T_m for Flat formation. The T_m decreases with the increase of d from 0.2m to 0.6m. Compared with Trefoil formation in the same buried depth, the T_m of Flat formation is lower and the change of d will significantly affect the heat dissipation of Flat formation. The T_m of Flat formation dropped 7.4K, 7.6K and 7.7K in $h=1m$, $h=1.25m$ and $h=1.5m$, respectively. In addition, the power cable system installed by Flat formation will achieve a lower T_m compared with Trefoil formation, which will be in a more secure loading environment.

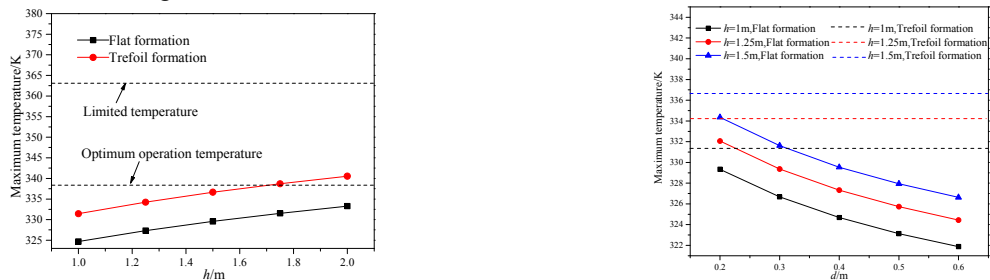


Fig. 3. (a) The changing of T_m under different h ; (b) The effect of cable spacing to T_m in Flat formation

3.3. The effect of thermal conductivity

Considering the thermal conductivity of actual soil dependent on the temperature, the dry zone will generate around the power cable system, which leads to deterioration of heat dissipation. Dry zone may result in a considerable increase in temperature. Compared the results of Fig. 4 with Fig. 3, it can be found that the temperature-dependent thermal conductivity corresponding to actual application will lead to the T_m increase rapidly for two formations, the results calculated with the constant thermal conductivity will be conservative. The computational results shown in Fig. 4 considering two different models of heat dissipation ($k(T) \neq \text{const}$, with FTB and $k(T) \neq \text{const}$, without FTB) are discussed. Compared them one can be observed that the T_m of Trefoil formation ($k(T) \neq \text{const}$, without FTB) is in a dangerous condition from $h=1m$ to $h=2m$. What's more, when the buried depth exceeds the 1.8m, the T_m will exceed the limited temperature which may lead to the transmission line malfunction.

The optimum operation temperature is specified as 338K, which will ensure the power transmission line security and economic. Therefore, the deepest buried depth cannot exceed 1.7m for the Trefoil formation and the suggested buried depth of the Flat formation can exceed 1.5m, which will achieve the maximum possible benefit in the 1145A loading current.

The process of heat transfer for power cable system needs to be enhanced, especially for the zone nearby the power transmission line. The T_m is significantly lower when the FTB is applied instead of the NS in the bottom layer. The range of T_m is in a reasonable and reliable region, which is from 319.8K to 332.2K for Flat formation and 323.3K to 337.4K for Trefoil formation. Compared with the installing the power cable system in the NS directly, the increase of T_m is slower. The using of FTB will ensure the system in a security condition, T_m for two kinds of formation are lower than optimum operation temperature. Hence, it means that the power system applied the FTB can have a larger load capacity in the same condition.

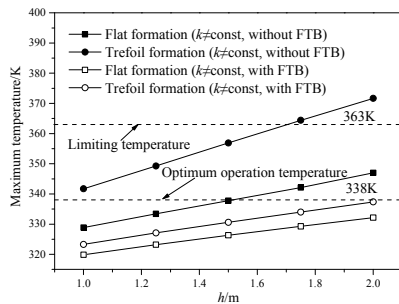


Fig. 4. Comparison of maximum temperature between two formations

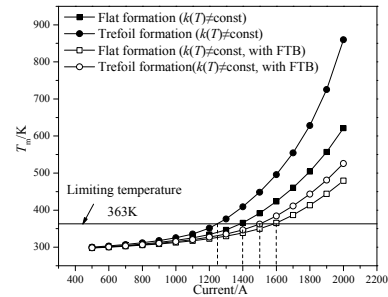


Fig. 5. The analysis of ampacity

3.4. Analyses of ampacity and distribution of thermal field

Ampacity is an important indicator in the power transmission. Fig. 5 shows the changes of T_m with the varied current for Flat formation ($k(T) \neq \text{const}$ without FTB, $k(T) \neq \text{const}$ with FTB) and Trefoil formation ($k(T) \neq \text{const}$ without FTB, $k(T) \neq \text{const}$ with FTB). When consider the temperature-dependent thermal conductivity, the ampacities of Flat formation ($k(T) \neq \text{const}$) and Trefoil formation ($k(T) \neq \text{const}$) are 1396.4A and 1248.5A respectively. The ampacities are 1599.3A and 1500.9A for Flat formation and Trefoil formation when use the FTB instead of the native soil layer in the proximity of cable line. It shows that the use of FTB will effectively improve the ampacity of power cable.

FTB does the significant cable conduct temperature drop shown in Fig. 6. Compared Fig. 6(a) and Fig. 6(b), Fig. 6(b) and Fig. 6(d), the Flat formation has a better heat dissipation than Trefoil formation for a wider arrangement. From Fig. 6(a) and Fig. 6(c), Fig. 6(b) and Fig. 6(d), the comparisons are observed that use of FTB leads to lower temperature distributions in the region of the layer nearby power cable. Moreover, a more uniform temperature field with the bottom layer can be observed when the FTB is used instead of NS. In addition, the T_m is in the middle cable in the Flat formation both with FTB and without FTB. As for Trefoil formation, three cables are almost in same temperature, which are in a high level. The effect of using FTB in Trefoil formation is better than it in Flat formation. Therefore, the application of FTB results in a significantly lower temperature in the underground power cable system. Flat formation will have a more suitable temperature for a security loading condition.

4. Conclusions

In this paper, the heat transfer of the two formations (Flat and Trefoil) for buried power cable system are discussed, the influences of geometrical configuration and temperature-dependent thermal conductivity on ampacity in different formations are researched, the main conclusions are drawn as follows:

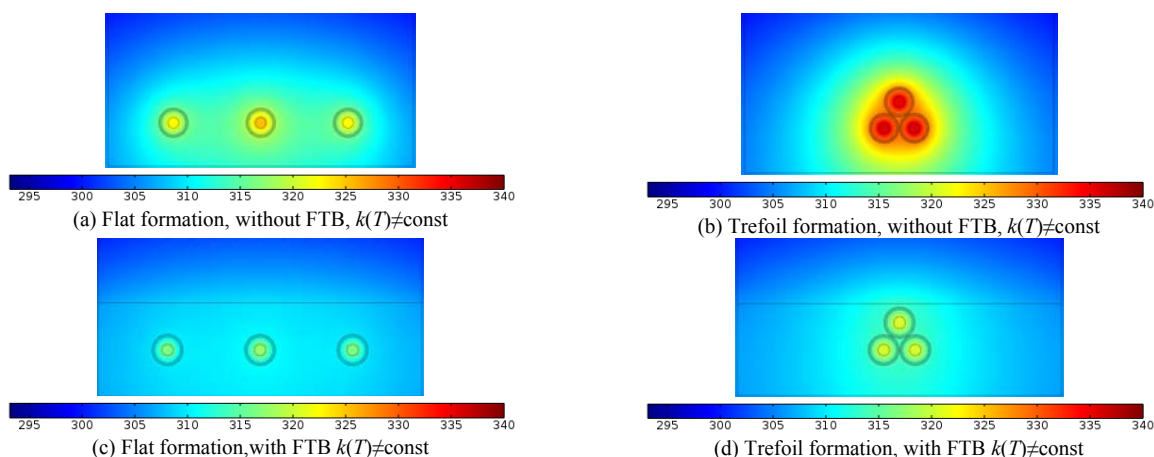


Fig. 6. The distributions of thermal field for two formations in layer nearby the power cables

(1) With the increase of buried depth, the maximum temperature of underground power cable system gradually increases, the reduction of spacing of cable in Flat formation will lead to the increase of temperature.

(2) The computational result with the condition of temperature-dependent thermal conductivity corresponding to actual application will lead to the increase of T_m compared with the constant thermal conductivity, which may cause the transmission line malfunction and the ampacity will be lower. The use of FTB can enhance the heat dissipation of power cable system, reduce the temperature. Therefore, a higher ampacity will be achieved in a security operation condition.

(3) The highest temperature values are obtained in the middle cable for the Flat formation both with FTB and without FTB. As for Trefoil formation, three cables are almost in same temperature, which are in a high level. The effect of the FTB is greater in Trefoil formation than Flat formation.

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